

App Dev-1 Project Report

1. Student Details

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About Me: I am a student at IIT Madras BS Degree with an interest in learning new technologies in my day to day life.

2. Project Details

Project Title: Trekking App

Problem Statement:

To design and build a web application that streamlines trek coordination by providing dedicated dashboards for Admins, Staff, and Trekkers to manage bookings, approvals, and trek logistics.

Approach:

The application is built using the Flask framework, utilizing a modular design that ensures maintainability and scalability. By implementing a role-based architecture, the system provides distinct, tailored experiences for Admins, Trek Staff, and Trekkers. It facilitates seamless administrative control over trek logistics while providing users with a robust, streamlined booking process.

3. AI/LLM Declaration

AI tools (Gemini) were employed to assist with Bootstrap 5 styling, debugging support, and the implementation of security protocols like password hashing. This accounts for roughly 20–25% of the development effort. I manually authored all foundational logic, database integration, and feature implementation to ensure project integrity.

4. Technologies and Frameworks Used

Technology/Library	Purpose
Flask	Core backend web framework
SQLAlchemy	ORM for SQLite database
Jinja2	Templating engine for rendering dynamic HTML pages
Bootstrap5	Frontend styling and responsive design
Session	User authentication and session management
SQLite	Lightweight local database for storing user data
Werkzeug	Security utility for secure password hashing

5. Database Schema / ER Diagram

Tables:

1. User – stores user details (id,name,email,password,phone no. , role, status,creating at)
2. Trek – stores trek details(id,name,location,difficulty,duration , available slots, max slots,start date,end date,status, assigned staff id, meeting point,description,cost per person)
3. Booking – stores booking details of trekker (id, user id, trek id, booking date,status)

Relationships:

- One-to-many → User → Booking
- One-to-many → Trek → Booking
- One-to-many → User → Trek



6. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for user, admin, staff , auth routes
- **/templates** – Jinja2 HTML templates
- **Requirements.txt** – For installation packages

Implemented Features:

Feature Category	Implementation Details
Authentication	Secure hashing, Session-based RBAC, Account approval logic
Admin Control	CRUD operations, Audit logs, User/Trek management
Staff Tools	Assigned trek filtering, Real-time slot & roster management
User Experience	Responsive UI, Booking history, Search & filter engine

7. Video Presentation:

Drive Link:

https://drive.google.com/file/d/1_5jbFacrARexwsCrzjXy4kD0iqAlZnGW/view?usp=drivesdk